

Demand

ECON 601, Lecture 2 in context

What Lecture 1 hands over, what Lecture 2 does with it, and where Lecture 3 takes it

How this is organised. Not section by section. The document follows the four questions Lecture 2 asks, in the order it asks them. Lecture 1 material appears where Lecture 2 needs it, flagged ← **L1**: like this. Lecture 3 appears as the destination, flagged → **L3**: like this. Page references point to the lecture PDFs; RS1–RS3 are the review session solutions; J&R is Jehle and Reny, 3rd edition.

Contents

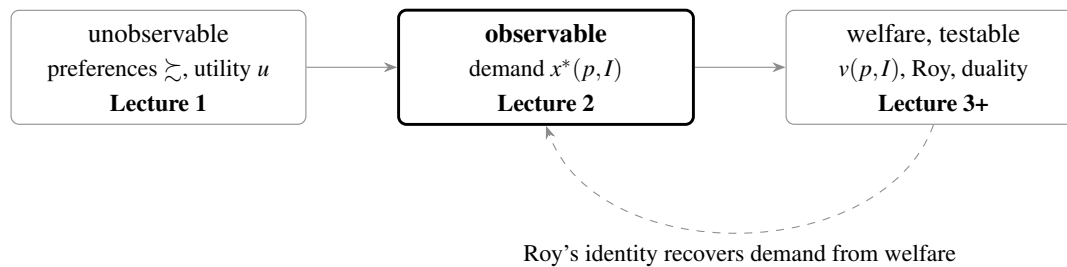
1	Why any of this exists	2
2	What Lecture 1 hands over	3
2.1	The delivery manifest	3
2.2	Two facts from Lecture 1 that shape every answer	3
2.3	Quasi-concavity, since Q2 runs on it	3
2.4	The four special functions are the four cases	4
3	Q1: does a best affordable bundle exist?	5
3.1	Why the question is not trivial	5
3.2	The budget set	5
3.3	Why that settles the question	5
3.4	Which hypothesis did what	5
4	Q2: is the answer one bundle or many?	7
4.1	Why existence did not settle this	7
4.2	The two results, and the single character that separates them	7
4.3	The mirror-image error	7
5	Q3: what is true regardless of who the consumer is?	9
5.1	Why this is the section that matters empirically	9
5.2	Property 1: demand is HD0 in (p, I)	9
5.3	The detour Lecture 2 takes here, and why	9
5.4	Property 2: the budget binds	10
6	Q4: how do you compute it?	12
6.1	The problem ordinary calculus cannot handle	12
6.2	The Lagrangian	12
6.3	Kuhn-Tucker, built from one variable	12
6.4	Tangency	14
6.5	The whole apparatus on one function	14

7	Where this goes: Lecture 3 and beyond	16
7.1	The move Lecture 3 makes	16
7.2	What carries over, and what does not	16
7.3	The envelope theorem, which is the actual tool	16
7.4	Roy's identity, and why the sequence was built this way	17
7.5	What comes next	17
8	Exercises, grouped by the question they answer	19
8.1	On Q1	19
8.2	On Q2	19
8.3	On Q3	19
8.4	On Q4	19
8.5	If you only work six	19
8.6	Beyond the review sessions	19
9	Reference card	21
9.1	The four questions and their answers	21
9.2	Core objects and identities	21
9.3	Error checks before submitting	21
9.4	Counterexamples to have ready	22

1 Why any of this exists

Preferences are invisible. Nobody can observe a ranking inside someone's head, and no survey recovers one reliably. Prices, incomes, and quantities purchased are visible, and governments and firms collect them constantly.

The entire construction is a bridge between those two facts. Lecture 1 builds an unobservable object, a utility function, from assumptions about an unobservable ranking. Lecture 2 pushes that object through a budget constraint to produce something observable: what the consumer buys. Lecture 3 then runs the bridge in the other direction, recovering what the consumer buys from how their welfare responds to prices, with no need to know the utility function at all.



Lecture 2 is the middle of that bridge, which is why it carries most of the weight. It answers four questions, and it answers them in this order for a reason: each question costs one more assumption than the last.

	Question	Assumption it costs	Where answered
Q1	Does a best affordable bundle exist at all?	continuity of u ; $p \gg 0$	Section 3
Q2	Is it one bundle or many?	curvature of u	Section 4
Q3	What is true of it no matter who the consumer is?	monotonicity	Section 5
Q4	How do you actually compute it?	differentiability	Section 6

That ordering is the single most useful thing to carry. When a later step confuses you, the question to ask is almost always: *which assumption is this step spending?* The four families are:

Family	Hypotheses	What it controls, and only this
Regularity	continuity of u , compactness of B	whether the maximum is attained, and how the <i>value</i> behaves
Curvature	quasi-concavity, strict quasi-concavity	how many <i>solutions</i> , and the shape of the solution set
Monotonicity	u strictly increasing in some good	<i>where</i> the solution sits, in particular whether the budget binds
Smoothness	differentiability	whether calculus is available

The mistake this prevents. Expecting a claim to deliver something from a different family. Continuity buys existence and says nothing about uniqueness. Strict quasi-concavity buys uniqueness and says nothing about whether income is spent. Both errors are common and both are avoided by naming the family first.

2 What Lecture 1 hands over

Lecture 1 is not a separate subject. It is the supplier. Everything in it that Lecture 2 uses fits on this page; the rest is background you can leave until revision.

2.1 The delivery manifest

Lecture 1 gives	Becomes	Used in Lecture 2 for
A1, A2 completeness and transitivity	a ranking	the problem being well posed at all
A3 continuity	a continuous u (Debreu)	Q1 : existence of x^*
A4 monotonicity	u increasing	Q3 : the budget binds
A5 convexity	u quasi-concave	Q2 : the solution set is convex
A5' strict convexity	u strictly quasi-concave	Q2 : the solution is unique
X closed, convex, $0 \in X$ (RS1 Q1.2)	properties of the domain	Q1 : B is closed and convex
Prop 1.1 invariance	u is ordinal	Q4 : a check on every answer
MRS and its invariance	the right marginal object	Q4 : the tangency condition

2.2 Two facts from Lecture 1 that shape every answer

Utility is ordinal (L1 Prop 1.1, p.6). If u represents $\succsim$ and h is strictly increasing, then $h[u]$ represents the same preferences. So infinitely many utility functions describe one consumer.

This is not philosophy; it is a working constraint. Any feature of your demand answer that depends on which representation you picked is an *error*. When a multiplicative constant survives into your demand function, you have made a mistake, and you know it without rechecking a line of algebra.

Marginal utility is the wrong primitive; MRS is the right one (L1 p.17). Replacing u by $1000u$ multiplies every marginal utility by 1000. Marginal utility is therefore meaningless on its own. The ratio survives, because the scaling cancels:

$$MRS_{2,1} \Big|_{u_0} = \frac{\partial u / \partial x_1}{\partial u / \partial x_2} = \left| \frac{dx_2}{dx_1} \right|_{u_0},$$

and L1 p.19 confirms u and $h[u]$ share it. This is exactly why the optimality condition in Lecture 2 is written as an MRS equalling a price ratio and not as marginal utilities equalling anything. The theory is built out of the objects that survive the ordinality problem.

2.3 Quasi-concavity, since Q2 runs on it

Definition 1 (L1 Def 1.3, p.13). f is quasi-concave if for all x^1, x^2 and $t \in [0, 1]$, $f(tx^1 + (1-t)x^2) \geq \min\{f(x^1), f(x^2)\}$, and strictly quasi-concave if the inequality is strict for $x^1 \neq x^2$, $t \in (0, 1)$.

RS2 Q1.1 derives this definition instead of asserting it. Start from convex preferences, take $x^1 \succ x^2$, note that the better-than set of x^2 is convex so the combination lies in it, and translate back through the definition of u . The min appears because $u(x^2)$ is the smaller of the two values. Equivalently (L1 Thm 1.2), f is quasi-concave exactly when every better-than set $\{x : f(x) \geq K\}$ is convex.

Trap. Diminishing MRS in every pair does *not* give quasi-concavity once there are more than two goods. Counterexample (RS2 Ex 1.3): for $u(x) = x_1 + (x_2 x_3)^{2/3}$, the bundles (16, 0, 0) and (0, 8, 8) are both worth 16, while the combination (14, 1, 1) is worth 15. L1 Proposition 1.5 states the equivalence for two goods only.

2.4 The four special functions are the four cases

These are one family, CES, at three limits (L1 pp.19–20; the Cobb-Douglas limit is proved in RS2 Q2.2). They are in the course because between them they exhaust the answers Lecture 2 can give.

Function	Curvature	Answer to Q2	Why it is in the course
Cobb-Douglas	strictly quasi-concave	unique, interior	the case where every theorem applies
Linear	quasi-concave, not strictly	multiple, or a corner	breaks uniqueness; breaks continuity of demand in L3
Leontief	quasi-concave, not strictly	unique, at the kink	shows strictness is sufficient, not necessary
Max (L1 p.20)	not quasi-concave	extremes preferred	non-convex preferences, the RS1 Q1.6 trip

The Leontief row is the one people miss. Its solution is unique despite failing strict quasi-concavity, which proves the Lecture 2 uniqueness result is a one-way implication. It is also the case where calculus fails outright, since there is no derivative at the kink.

Background you can defer: Debreu's theorem and its proof, lexicographic preferences, the countability argument, and the concave-versus-quasi-concave examples. None of it is used by Lecture 2. Read it when the axioms-to-utility step starts to feel like a black box.

3 Q1: does a best affordable bundle exist?

3.1 Why the question is not trivial

It is easy to write down a maximisation and assume it has an answer. It need not. Two things can go wrong. The consumer might improve forever without arriving, which happens when the feasible set is unbounded. Or the objective might jump, so the supremum is approached but never reached.

Ruling both out is the job of this section, and everything after it silently assumes the job was done.

3.2 The budget set

Lecture 1 gave preferences but no scarcity: everything in X was imaginable. Scarcity enters here. In two goods, B is the triangle with corners at the origin, $(I/p_1, 0)$ and $(0, I/p_2)$.

Definition 2 (L2 p.2). $B = \{x \in X : p \cdot x \leq I\}$, with $p \gg 0$ and $I > 0$.

Proposition 1 (L2 p.2, left as an exercise). B is nonempty, convex and compact.

Proof. *Nonempty:* the origin lies in $X = \mathbb{R}_+^n$ and costs nothing, and $I > 0$. $\leftarrow$ **L1:** $0 \in X$ is one of the standing assumptions on the consumption set, RS1 Q1.2.

Convex: take $x^1, x^2 \in B$ and $t \in [0, 1]$. The combination stays in X because X is convex. By linearity of the dot product,

$$p \cdot (tx^1 + (1-t)x^2) = t(p \cdot x^1) + (1-t)(p \cdot x^2) \leq tI + (1-t)I = I.$$

Closed: $g(x) = p \cdot x$ is continuous and $B = X \cap g^{-1}((-\infty, I])$. The preimage of a closed set under a continuous map is closed, X is closed, and intersections of closed sets are closed.

Bounded: fix i . Every term in the expenditure sum is nonnegative, so one term is no larger than the whole:

$$p_i x_i \leq \sum_j p_j x_j = p \cdot x \leq I \quad \implies \quad 0 \leq x_i \leq \frac{I}{p_i}.$$

So B sits inside the box $\prod_i [0, I/p_i]$. Closed and bounded gives compact. □

3.3 Why that settles the question

Lecture 2, p.2 lists three facts about compactness without naming the theorem they amount to. Together they are Weierstrass: a continuous function on a compact set attains its maximum. Compactness of B stops the consumer escaping to infinity; continuity of u stops the maximum falling through a gap.

Definition 3 (UMP and Marshallian demand, L2 p.2).

$$\max_{x \in \mathbb{R}_+^n} u(x) \quad \text{s.t.} \quad p \cdot x \leq I, \quad \text{solution } x^*(p, I).$$

3.4 Which hypothesis did what

The convexity half of the proof used nothing about prices. The boundedness half used $p \gg 0$, and only there, to divide by p_i .

So if some $p_i = 0$, then B stays nonempty, convex and closed, but x_i is unbounded. B is no longer compact, Weierstrass no longer applies, and if u is strictly increasing in good i the UMP has *no solution at all*. This is the cleanest example in the course of one hypothesis doing exactly one job.

$\rightarrow$ **L3:** Indirect utility is defined as the maximised value, $v(p, I) = u(x^*(p, I))$. That definition is only meaningful because the maximum is attained. Lecture 3 never revisits this; it simply assumes Q1 was answered.

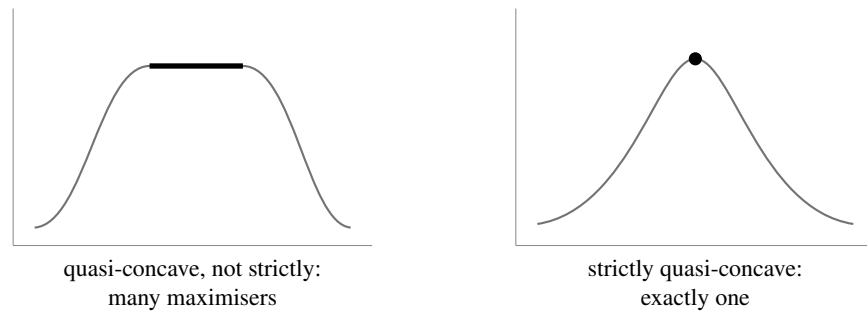
Carry the convexity result forward. The three-line convexity argument above gets reused verbatim three more times: in the uniqueness proof (Section 4), in the convex-solution-set proof (Section 4), and in the quasi-convexity proof of Lecture 3. In all four the move is identical: scale two inequalities by weights summing to one, add, and recognise the result.

4 Q2: is the answer one bundle or many?

4.1 Why existence did not settle this

These are different questions and they are secured by different features.

Existence asks whether the best value is reached, and is settled by regularity. Uniqueness asks whether that value is reached at one point or several, and regularity has nothing to say about it. Continuity restricts local jumps; it places no restriction at all on whether two distant bundles tie. A constant function is perfectly continuous and every point maximises it.



The lecture's own counterexample is on p.3: with $u = x_1 + x_2$, $p_1 = p_2 = 1$ and $I = 10$, the maximum is 10 and every bundle on the budget line attains it. The figures on pp.3 and 4 are the two panels above, drawn as indifference curves.

4.2 The two results, and the single character that separates them

Proposition 2 (L2 p.3; RS3 Q1.1). *If u is strictly quasi-concave then x^* is unique.*

Proof. Suppose distinct x^{J*} and x^{I*} both solve the UMP, so $u(x^{J*}) = u(x^{I*})$. Form $x^{t*} = tx^{J*} + (1-t)x^{I*}$ for $t \in (0, 1)$. $\leftarrow$ **L1:** Convexity of B from Q1 makes this feasible. Strict quasi-concavity gives $u(x^{t*}) > \min\{u(x^{J*}), u(x^{I*})\} = u(x^{J*})$, contradicting optimality. $\square$

Proposition 3 (L2 p.6). *If u is quasi-concave then the set of solutions is convex.*

Proof. Let v be the maximised value and take two solutions. The combination is feasible by the same appeal to convexity of B , and quasi-concavity gives $u(x^*) \geq \min\{v, v\} = v$. But v is the maximum over B , so $u(x^*) \leq v$ too. Hence $u(x^*) = v$ and the combination is itself a solution. $\square$

Read the two proofs side by side. Same setup, same combination, same appeal to convexity of B . The only difference is whether the key inequality is strict. Strict produces a contradiction, so the second solution cannot exist. Weak produces no contradiction and instead proves membership. One character flips the conclusion from “exactly one” to “possibly many, forming a convex set.”

Sufficient, not necessary. $\leftarrow$ **L1:** Leontief preferences are not strictly quasi-concave, yet the UMP solution is unique, at the kink. The proposition is a one-way implication. Failing its hypothesis tells you nothing.

4.3 The mirror-image error

Having just learned that strictness buys uniqueness, it is tempting to think it buys tidiness in general. It does not. Take the bliss-point utility

$$u(x, y) = -(x - x_0)^2 - (y - y_0)^2, \quad I > p_x x_0 + p_y y_0.$$

This is strictly quasi-concave, so the solution is unique, and continuous, so it exists. And the consumer leaves income unspent (RS2 Q2.3). Uniqueness bought one solution and said nothing about *where* that solution sits. That is Q3's job.

→ **L3:** The same split appears one level up. Continuity of u gives continuity of v but not of x^* ; Proposition 3.1 needs strict quasi-concavity for demand to be continuous (*L3 p.5*). Regularity governs the value, strictness governs the solution. With linear utility, $v = I / \min\{p_1, p_2\}$ glides smoothly while demand jumps from one axis to the other. Same logic, same counterexample, one lecture later.

5 Q3: what is true regardless of who the consumer is?

5.1 Why this is the section that matters empirically

Q1 and Q2 are housekeeping: they establish that the object exists and is well defined. They say nothing you could take to data, because they are statements about the model.

This section is different. Both results here hold for *every* consumer satisfying the axioms, whatever their tastes. That makes them the only claims so far that could be falsified by observing actual purchases, and it is why they are the empirical content of consumer theory.

5.2 Property 1: demand is HD0 in (p, I)

$$x^*(tp, tI) = t^0 x^*(p, I) = x^*(p, I) \quad \text{for all } t > 0$$

Why it must be true. Demand cannot see prices and income at all. It can only see the budget set. And scaling every price and income by the same factor draws the *identical* triangle, not a similar one. The consumer is solving a problem already solved.

Formally (L2 p.5), the objective $u(x)$ contains no prices or income, and the scaled constraint $tp \cdot x \leq tI$ divides by t back into the original. Same objective over the same feasible set gives the same solution. The proof needs nothing about the consumer, which is exactly the point.

Intuition. Scaling a triangle changes its side lengths but not its angles. Lengths are degree one, angles are degree zero. Demand is an angle. The concrete version is a currency redenomination: a country drops three zeros, a coffee goes from 5,000 to 5 and your salary from 4,000,000 to 4,000, and you buy the same coffee.

Three uses.

1. *It removes a variable.* Choosing $t = 1/I$ gives $x^*(p, I) = x^*(p/I, 1)$, so demand really depends on n numbers, prices measured in units of income. This is why general equilibrium normalises prices, and why only relative prices are ever determined.
2. *It is a free error check.* Scale your final expression and confirm the scaling cancels. This is the economist's version of checking units, and it catches algebra errors in seconds.
3. *It is a restriction you can estimate.* See the elasticity form below.

It is an assumption, not a tautology. HD0 is the formal statement that this consumer has no money illusion. Real people demonstrably respond to nominal magnitudes: they resist nominal wage cuts while accepting real cuts delivered by inflation, and they anchor on remembered nominal prices. The model was built to have this property. Knowing which step delivers it, the line where t divides out of the budget constraint, is what lets you say something sensible when model and behaviour disagree.

5.3 The detour Lecture 2 takes here, and why

To turn HD0 into a statement about derivatives, the lecture stops and introduces homogeneous functions (L2 pp.4–5). This looks like a digression and is not.

Definition 4 (L2 Def 2.1). f is homogeneous of degree k if $f(tx) = t^k f(x)$ for all x and all $t > 0$.

The crucial remark, made explicitly in RS3, is that this is an *identity*: it holds for every x and every t , so it may be differentiated with respect to either. Most people read the definition as a test to apply and never notice it is an equation to manipulate.

Theorem 1 (Euler, L2 Thm 2.1). For differentiable f , f is homogeneous of degree k if and only if $\sum_i \frac{\partial f}{\partial x_i} x_i = k f(x)$ for all x .

Why it is true, geometrically. Walk outward along the ray through a bundle and record the value, $g(t) = f(tx)$. Homogeneity says that along every such ray f is an ordinary power function. Differentiating that one-variable function two ways, directly and by the chain rule, and evaluating at $t = 1$, gives the theorem. The left side is the rate of change of f as you move radially outward from the origin; Euler says that radial rate is pinned down by the value and the degree, and nothing else.

What Euler is for. Homogeneity is a global claim about every scaling at once, which is awkward to check and impossible to differentiate. Calculus is local. Euler is the bridge between them, converting a global scaling property into one equation holding at each point, involving nothing but first derivatives.

The payoff, immediately. Apply it to demand with $k = 0$, remembering the arguments are the n prices *and* income:

$$\frac{\partial x_i^*}{\partial p_1} p_1 + \cdots + \frac{\partial x_i^*}{\partial p_n} p_n + \frac{\partial x_i^*}{\partial I} I = 0 \quad \Longleftrightarrow \quad \sum_j \varepsilon_{ij} + \eta_i = 0.$$

In words: all the price elasticities for good i and its income elasticity sum to exactly zero. Estimate a demand system on real data and you can test that. A definitional property became a falsifiable one, and Euler is what did it.

Trap. Always check *which* variables the homogeneity is in before writing the Euler sum. Demand is homogeneous in prices and income together, so the sum has $n + 1$ terms. A utility function homogeneous in quantities gives n terms. Wrong count is the most common error with this theorem.

The companion result (*RS3 Prop 1*) is that $\partial f / \partial x_i$ is homogeneous of degree $k - 1$. That is the bookkeeping which keeps the degrees consistent across the course.

Object	Degree	In	Where
Marshallian demand	0	prices and income	L2
Indirect utility	0	prices and income	L3
Expenditure function	1	prices	next
Hicksian demand	0	prices	next
Cost, profit functions	1	input prices	later in the sequence

5.4 Property 2: the budget binds

Proposition 4 (L2 p.6; RS3 Q3.2(b)). *If u is strictly increasing in some x_i , then $p \cdot x^* = I$.*

← **L1:** This is Axiom 4 cashing in. Monotonicity was introduced in Lecture 1 with the remark that it would “later ensure the use of one’s income.” This is later.

Proof. Suppose not. Any solution is feasible, so failure means $p \cdot x^* < I$. Let good i be one in which u is strictly increasing, and set

$$\delta = I - p \cdot x^* > 0, \quad \varepsilon = \frac{\delta}{p_i} > 0,$$

the division being legitimate because $p \gg 0$. Construct $\bar{x} = (x_1^*, \dots, x_i^* + \varepsilon, \dots, x_n^*)$. It is nonnegative, and

$$p \cdot \bar{x} = p \cdot x^* + p_i \varepsilon = p \cdot x^* + \delta = I,$$

so it is feasible. Since u is strictly increasing in good i and only that component rose, $u(\bar{x}) > u(x^*)$, contradicting optimality. $\square$

Two remarks. The lecture says “increase x by ε ” without specifying ε ; choosing the one that exactly exhausts the slack makes feasibility an equality instead of an estimate. And the hypothesis is deliberately weak: strictly increasing in *some* good, not all. You only need one good worth buying more of.

Why it is needed. Without it the first-order conditions give you ratios and nothing else. The binding budget is the extra equation that converts a ratio into quantities, which is why every worked example in Section 6 deduces it before solving.

→ **L3:** In Lecture 3 this result is what makes $\partial v / \partial I = \lambda^* > 0$. Roy's identity divides by that derivative, so the strict positivity is load-bearing, not decoration. It also signs $\partial v / \partial p_i = -\lambda^* x_i^* < 0$. Further ahead, the mirror result in expenditure minimisation, that the utility constraint binds, is what makes duality close.

6 Q4: how do you compute it?

6.1 The problem ordinary calculus cannot handle

Calculus finds a maximum by setting derivatives to zero, which works because at the top of an unconstrained hill the ground is flat. Consumer problems break this. The unconstrained peak of a utility function is at infinite consumption. The point you actually reach sits on the edge of the budget set, and there the derivative of utility is emphatically not zero. You are still going uphill; you have run out of money.

Everything in this section is machinery for getting calculus back.

6.2 The Lagrangian

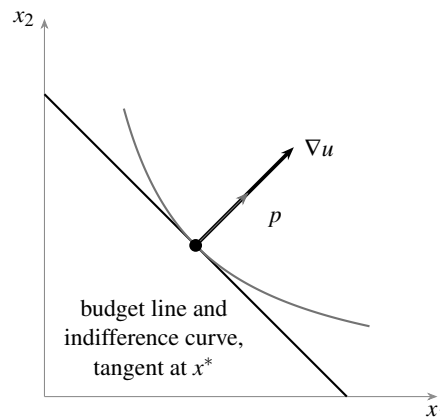
Definition 5 (L2 p.7).

$$\mathcal{L}(x, \lambda) = u(x) + \lambda [I - p \cdot x], \quad \max_x \min_{\lambda} \mathcal{L} \text{ s.t. } x \geq 0, \lambda \geq 0.$$

The bracket is leftover income: zero on the budget line, positive inside the budget set, negative if you overspend. Read λ as a price on money. Every dollar left over is credited at rate λ , every dollar of overspending charged at rate λ . You are now free to buy anything at all, because overspending is priced instead of forbidden. The constraint has been folded into the objective.

Why max then min. The inner minimisation is what enforces the constraint. Propose an unaffordable bundle: the bracket is negative, so minimising over $\lambda \geq 0$ drives λ up without bound and the expression to $-\infty$. Cheating is made infinitely unattractive. Propose a bundle with money left over: the bracket is positive, so the minimisation sets $\lambda = 0$, killing the term and leaving plain utility. The penalty is thus set automatically at exactly the right severity, which is why the figure on p.7 shows a saddle point.

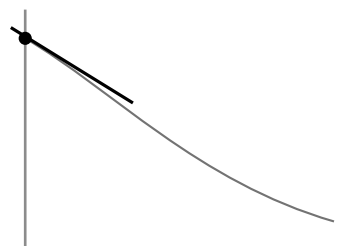
The geometry. Setting the quantity derivatives to zero gives $\nabla u(x^*) = \lambda^* p$. The gradient is the direction of steepest increase in utility; p is perpendicular to the budget line. So at the optimum, uphill points straight out of the budget set. That is precisely the condition for being stuck: if the gradient had any component running *along* the budget line you could slide that way, stay affordable, and gain. The multiplier is only the proportionality constant between two vectors that turned out to be parallel.



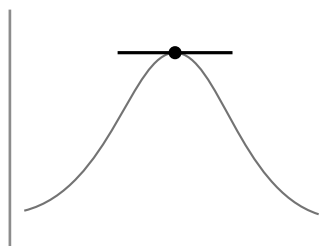
6.3 Kuhn-Tucker, built from one variable

The conditions on p.7 look forbidding. They are one idea from single-variable calculus, repeated once per good. Strip away utility, prices and vectors.

Step 1: maximising when the variable cannot go negative. Consider $\max_x f(x)$ subject to $x \geq 0$. Two things can happen, and they are the two pictures on L2 p.8.



Corner. Best at the wall,
slope still negative



Interior. Best in the middle,
slope zero

For $f(x) = 10x - x^2$ the peak is at five and the slope there is zero. For $f(x) = -3x - x^2$ the slope at zero is -3 and only worsens, so the best you can do is sit at zero, and the slope at the optimum is *not* zero. You would love to move left and climb, but negative quantities are forbidden. You are pinned against a wall.

Step 2: one statement covering both. Could the slope at the optimum be positive? No: you could then step right, stay legal, and do better. That rules out one of three possibilities and leaves two, which is the first line: $f'(x^*) \leq 0$. Strictly negative means pinned at the wall; zero means a genuine hilltop.

Step 3: the product is an “or” written as an equation. Either the quantity is zero or the slope is zero. Two numbers multiply to zero exactly when one of them is zero, so that “or” compresses to $f'(x^*) \cdot x^* = 0$. That is all complementary slackness is. In the first example the quantity is five and the slope is zero; in the second the quantity is zero and the slope is -3 . Both satisfy it, and nothing else does.

Step 4: put $\mathcal{L}$ in the role of f . Holding λ fixed, the consumer’s problem is exactly this shape with several quantities instead of one. Apply Steps 2 and 3 good by good.

Definition 6 (Kuhn-Tucker conditions, L2 p.7).

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial u}{\partial x_i} - \lambda p_i \leq 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - p \cdot x^* \geq 0,$$

$$x^* \geq 0,$$

$$\frac{\partial \mathcal{L}}{\partial x_i} x_i^* = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} \lambda^* = 0,$$

$$\lambda^* \geq 0.$$

Step 5: the multiplier line is the same idea, flipped. The multiplier also cannot go negative, but $\mathcal{L}$ is *minimised* over it, so both signs flip. Its derivative is leftover income, so the inequality is only the budget constraint reappearing, and the product is again an “or”: either there is no leftover, so the budget binds, or the multiplier is zero, so an extra dollar is worth nothing.

Line in the notes	What it says
derivative in good i at most zero that derivative times the quantity is zero	you never leave a good where buying more would help either you buy none of good i , or it is exactly worth its price
leftover income at least zero leftover times the multiplier is zero	you did not overspend either you spend everything, or money is worthless to you
quantities and multiplier at least zero	you cannot buy negative amounts, and the penalty is not a subsidy

Step 6: watch it work. Take the quasi-linear example on L2 p.9, $u = x_1 + \ln x_2$, with both prices one. With $I = 10$ the answer is $x_1^* = 9$, $x_2^* = 1$, $\lambda^* = 1$: both first-order conditions hold with equality, both quantities are positive, leftover is zero. Now drop income to $1/2$. The answer becomes $x_1^* = 0$, $x_2^* = 1/2$, $\lambda^* = 2$, and

$$\frac{\partial u}{\partial x_1} - \lambda p_1 = 1 - 2 = -1 < 0, \quad x_1^* = 0,$$

so the product is still zero. Good one is not worth its price at this income, so you buy none and sit pinned at the wall with a negative slope. Setting that derivative equal to zero, as ordinary calculus suggests, would have produced a negative quantity and a wrong answer. That case is the whole reason the apparatus exists.

6.4 Tangency

If $x_i^* > 0$ and $x_j^* > 0$, both conditions hold with equality and dividing cancels λ^* :

$$\underbrace{\frac{\partial u / \partial x_i}{\partial u / \partial x_j}}_{MRS_{j,i}} = \underbrace{\frac{p_i}{p_j}}_{\text{price ratio}}.$$

← **L1:** This is why Lecture 1 built the theory on MRS. Marginal utilities individually are meaningless under monotone transformation; their ratio is invariant, so this equation says something about the consumer and not about the arbitrary choice of u .

The intuition on p.10 is the clearest sentence in the lecture. If $MRS_{j,i} = 2$ while the price ratio is one, you would privately trade one unit of i for two of j while the market trades them one for one. You value j less than everyone else does, so sell j and buy i until the ratios meet.

Rearranged one good at a time, the same conditions read $(\partial u / \partial x_i) / p_i = \lambda^*$: extra utility per dollar must be equal across every good you buy, because otherwise you would move a dollar from the low-return good to the high-return one. That common rate is what an extra dollar of income is worth, which is why λ^* is called the marginal utility of income.

6.5 The whole apparatus on one function

Cobb-Douglas is worth doing in full because it is the only one of the four special functions that satisfies every hypothesis in the strict form, so every result from Q1 to Q4 applies at once. ← **L1:** It is also not an arbitrary example: it is the CES family at the limit $\rho \rightarrow 0$, proved in RS2 Q2.2.

Each axiom becomes a visible feature of the answer.

From Lecture 1	Why Cobb-Douglas has it	What it produces in the answer
A1, A2	free: comparing values of a real function inherits both from $\geq$ on $\mathbb{R}$	the problem is well posed
A3 continuity	u continuous on $\mathbb{R}_+^n$	Q1: a solution exists
A5' strict convexity	strictly quasi-concave; L1 p.15 verifies it for xy via the hyperbola	Q2: the solution is unique
A4 monotonicity	increasing in each good	Q3: the budget binds
Prop 1.1 invariance	$A \cdot (\cdot)$ is a strictly increasing transformation	A must vanish from the answer
Prop 1.4 diminishing MRS	$MRS_{2,1} = \frac{\alpha}{1-\alpha} \frac{x_2}{x_1}$	Q4: the tangency condition is valid

The derivation (RS3 Q3.4). For $u = Ax_1^\alpha x_2^{1-\alpha}$, assume an interior solution so both first-order conditions hold with equality. Dividing one by the other removes λ and A at once and leaves the tangency condition, hence

$$\frac{\alpha}{1-\alpha} \cdot \frac{x_2}{x_1} = \frac{p_1}{p_2} \implies \frac{p_2 x_2}{p_1 x_1} = \frac{1-\alpha}{\alpha}.$$

Now invoke Q3: the budget binds. Substituting gives $p_1 x_1 / \alpha = I$ and therefore

$$x_1^* = \frac{\alpha I}{p_1}, \quad x_2^* = \frac{(1 - \alpha)I}{p_2}.$$

Why the exponents are expenditure shares. The boxed display above is the reason, and it is worth seeing that it has nothing to do with Cobb-Douglas being special. Tangency forces the ratio of spending to equal the ratio of exponents; the binding budget then splits income in exactly those proportions. The famous constant-share result is Lecture 1's MRS meeting Lecture 2's tangency condition, and nothing more.

The audit, which is the real lesson. Check the answer against Q1 to Q4 before moving on.

1. Scale (p, I) by t : the t cancels top and bottom, confirming HD0.
2. Add up: $\alpha I + (1 - \alpha)I = I$, confirming the budget binds.
3. The constant A vanished, as ordinality requires. RS3 Q3.5 asks you to redo the whole thing with $\ln u$ and verify identical demands.
4. Both quantities are strictly positive for any $p \gg 0$ and $I > 0$, so the interiority assumption was self-consistent.

Doing this is worth more under exam pressure than being fast, because it catches your own algebra errors in seconds. It also gives you one function you know cold, so that later general claims can be tested immediately. When a claim fails even for Cobb-Douglas it is false; when it holds for Cobb-Douglas but you suspect strictness matters, linear utility is the second thing to try.

→ **L3:** The Lagrangian you just built is reused immediately, but differentiated with respect to the *parameters* instead of the quantities. That is the envelope theorem, and it produces $\partial v / \partial I = \lambda^*$ and $\partial v / \partial p_i = -\lambda^* x_i^*$, whose ratio is Roy's identity. Lecture 3 does not introduce new machinery here. It points the same machinery in a different direction.

7 Where this goes: Lecture 3 and beyond

7.1 The move Lecture 3 makes

Lecture 2 produced $x^*(p, I)$, a function from parameters to quantities. Lecture 3 asks a different question about the same problem: not what you buy, but how well off you end up.

Definition 7 (Indirect utility, L3 p.2). $v(p, I) = u(x^*(p, I))$, the maximised value of the UMP.

Why bother? Because v is expressed entirely in observable, policy-relevant variables. RS3 puts it well: utility functions let you compare bundles conveniently, and indirect utility lets you compare (p, I) pairs conveniently. Taxes, subsidies, inflation and price shocks are all changes in (p, I) , so v is the object welfare analysis actually uses.

7.2 What carries over, and what does not

Property of v	Where it comes from	Same as demand?
HD0 in (p, I)	the budget set is unchanged	yes, identical argument
decreasing in p_i , increasing in I	scaling the feasible set up or down	new
quasi-convex in (p, I)	budget-set containment	new, and the least obvious
continuous in (p, I)	Berge	value only, <i>not</i> demand
invariant to monotone transforms of u	—	no : demand is, v is not

That last row is a favourite exam target. For the same preferences, different representations of u give different v values, though the ordering of those values is unchanged.

The space trap. Lecture 1 gave quasi-concavity of u over *commodity* space via *better-than* sets. Lecture 3 gives quasi-convexity of v over *price-income* space via *worse-than* sets. Same machinery, different object, different space, opposite direction. And the result holds whether or not u is quasi-concave, because the proof never touches the shape of u : it shows $B(p^t, I^t) \subseteq B(p^1, I^1) \cup B(p^2, I^2)$ using the same scale-and-add move you first met proving B convex in Section 3.

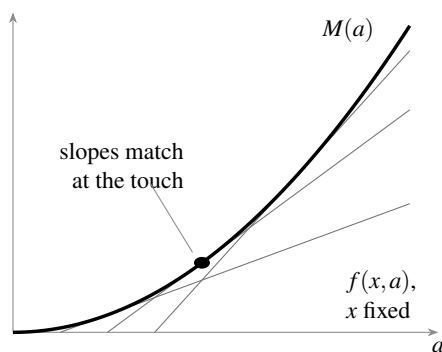
7.3 The envelope theorem, which is the actual tool

Everything above is description. This is the instrument, and it is what makes the rest of the sequence cheap.

In one line. When you are already at the top, moving does not help you, so you may ignore the fact that you will move.

The picture. Stand at the highest point on a hill and let someone tilt the landscape. Your height changes for two reasons: the ground under your feet moved, and you will shuffle to the new summit. But you were at the summit, so the ground around you was flat, and shuffling buys almost nothing. To first order the new height is the old spot with the ground moved under it. The indirect effect through re-optimising vanishes, because at an optimum the slope is zero.

Where the name comes from. Fix x and ask what $f(x, a)$ does as a changes: one curve. Do that for every x and you get a family. $M(a) = \max_x f(x, a)$ is the highest one at each a , so it sits on top of the family and touches whichever curve is best there. At the touching point the two are tangent, so their slopes agree. The slope of M is what you want; the slope of the fixed- x curve is easy, because x is not moving.



Statement and check (L3 pp.5–7). The direct method gives

$$M'(a) = \underbrace{f_x(x^*, a)}_{=0 \text{ by the FOC}} \cdot x^{*'}(a) + f_a(x^*(a), a) = f_a(x^*(a), a),$$

so the envelope method skips to the end: differentiate f with respect to a treating x as a parameter, then evaluate at $x^*(a)$. Test it on $f(x, a) = ax - x^2$. Directly, $x^*(a) = a/2$, $M(a) = a^2/4$, $M'(a) = a/2$. By envelope, hold x fixed and differentiate, getting just x , then evaluate at $a/2$. Same answer, and $x^{*'}(a)$ was never computed. Each straight line in the figure is this f at fixed x and its slope is that x , so the slope of M is simply the x that is optimal there.

← **L1:** You have already met the multiplier as the marginal value of a constraint: in the small worked example of Section 6, $\lambda^* = 5$ equalled $M'(10)$ exactly. That was the envelope theorem before it was named.

7.4 Roy's identity, and why the sequence was built this way

Lemma 1 (L3 Lemma 3.1). If u is differentiable, strictly increasing and strictly quasi-concave, then

$$x_i^*(p, I) = -\frac{\partial v / \partial p_i}{\partial v / \partial I}.$$

Proof. Write v as the saddle value of the Lagrangian and apply the envelope theorem to each parameter, holding x and λ fixed:

$$\frac{\partial v}{\partial I} = \lambda^* > 0, \quad \frac{\partial v}{\partial p_i} = -\lambda^* x_i^* < 0.$$

Dividing cancels λ^* and leaves x_i^* . □

The entire content is that you were allowed to hold x^* fixed while differentiating. The direct method would require a chain-rule expansion across every good. Note also which hypotheses are load-bearing: strict monotonicity is what makes the denominator strictly positive, via the binding-budget result of Q3.

Why this closes the loop. Return to Section 1. Utility is invisible; prices, incomes and quantities are not. Roy's identity recovers what the consumer buys purely from how their welfare responds to prices and income, with no knowledge of u . Every question Lecture 2 answered was a step toward being able to write that formula down.

7.5 What comes next

Lecture 3 ends by noting that $\partial x_i^* / \partial p_i$ is complicated because several effects are mixed together, and that an intermediate, more abstract problem is needed first. That problem is expenditure minimisation, and it opens the second half of the chapter:

- the expenditure function $e(p, u)$ and Hicksian demand $h(p, u)$, the mirror images of v and x^* (J&R 1.4.2);
- the duality relations tying all four objects together (J&R 1.4.3);

- the Slutsky decomposition into income and substitution effects, which is the answer to the question Lecture 3 leaves hanging (*J&R 1.5.2*);
- elasticity relations, the fuller version of the restriction Euler gave you in Q3 (*J&R 1.5.3*).

The pattern to expect: the mirror problem has a mirror binding constraint, mirror homogeneity degrees, and its own envelope result, Shephard's lemma, standing where Roy's identity stands here.

8 Exercises, grouped by the question they answer

Four of the results proved above are left as “Proof. Exercise.” in the lecture notes. Yildirim writes out the proofs he considers instructive to read and leaves the ones he considers instructive to do, which makes them the closest thing to a stated exam preview available. The proofs are in Sections 3 to 6; what follows is what each one is *for*.

8.1 On Q1

B is nonempty, convex, compact (L2 p.2). The only result in Lecture 2 whose failure would be invisible to you, since everything after p.2 silently assumes a solution exists. It installs the cheapest and most-used skill in the course: write down what a definition literally asserts and check it. Its convexity half is consumed three more times.

You have it when: you can say which of the four properties fails if a price is zero, and what goes wrong downstream.

8.2 On Q2

Strict quasi-concavity gives uniqueness (L2 p.3; RS3 Q1.1; again as RS3 Q3.2(a)). Anything written out three times is not subtle signalling. It installs proof by contradiction with a *constructed witness*, which is where most people stall. The same skeleton drives the quasi-convexity proof in Lecture 3.

You have it when: you can state without looking which step uses convexity of B and which uses strictness.

Quasi-concavity gives a convex solution set (L2 p.6). Low value alone, high value as a contrast with the one above. Do it second, and read them side by side.

You have it when: you can predict, before any algebra, which conclusion a weak hypothesis will produce.

Also: RS3 Q3.3 (convex but not strictly: existence survives, uniqueness does not) and RS2 Q1.1, Q1.2, Ex 1.3 on quasi-concavity itself. RS2 Ex 1.3 is the best single question in the review sessions.

8.3 On Q3

HD k implies Euler (L2 p.4; RS3 Q1.2). The only pure technique on the list, and the statement matters far less than the move: notice the definition is an identity, then differentiate it. The payoff compounds, since degree zero gives the demand restriction and degree one gives product exhaustion later in the sequence.

You have it when: you can say why setting $t = 1$ at the end is legitimate, and write the Euler sum for demand with the right number of terms without counting.

The budget binds (L2 p.6; RS3 Q3.2(b)). Read it with RS2 Q2.3, which asks the true-or-false version and answers false using the bliss point. The pair shows exactly which assumption carries the result. Add RS3 Q3.6 ($u = x_1$) for the corner case.

8.4 On Q4

Cobb-Douglas, then the log transform (RS3 Q3.4, Q3.5). The derivation is not the reason; the audit is. And it gives you one function you know cold to test later claims against.

You have it when: you can derive the demands without writing the Lagrangian, going straight from tangency plus a binding budget.

8.5 If you only work six

RS2 Q1.1, RS2 Ex 1.3, RS3 Q1.1, RS3 Q1.2, the RS2 Q2.3 with RS3 Q3.2(b) pair, and RS3 Q3.4 with Q3.5. That covers axiom-to-function translation, the main counterexample, uniqueness, the Euler technique, what makes the budget bind, and ordinality.

8.6 Beyond the review sessions

The review sessions work textbook problems and label them: RS1 Q1.2–Q1.6 are J&R Exercises 1.1–1.6, and RS3 Q3.1–Q3.6 are Exercises 1.12, 1.16, 1.17, 1.20, 1.21 and 1.26. They cover roughly a third of Section 1.6. The rest are

the same type, on the same material, from the same source your TAs draw from, without posted solutions. On Lectures 1 and 2: Exercises 1.5, 1.7, 1.8, 1.9, 1.10 and 1.13–1.15. Exercise 1.7 is the direct companion to the two Q2 proofs.

9 Reference card

9.1 The four questions and their answers

	Question	Answer	Needs
Q1	does x^* exist?	yes	u continuous, B compact, so $p \gg 0$
Q2	is it unique?	yes if strictly quasi-concave; otherwise the solution set is convex	curvature
Q3	what always holds?	HD0 in (p, I) ; $p \cdot x^* = I$	monotonicity for the second
Q4	how to compute?	Lagrangian, Kuhn-Tucker, $MRS = \text{price ratio}$	differentiability

9.2 Core objects and identities

$B = \{x \in X : p \cdot x \leq I\}$	nonempty, convex, compact
$x^*(p, I) = \arg \max_{x \in B} u(x)$	HD0 in (p, I)
$v(p, I) = u(x^*(p, I))$	HD0, decreasing in p_i , increasing in I , quasi-convex, continuous
$\mathcal{L} = u(x) + \lambda [I - p \cdot x]$	$\lambda^* = \partial v / \partial I > 0$

Euler:	$\sum_i \frac{\partial f}{\partial x_i} x_i = k f(x)$	Tangency:	$MRS_{j,i} = \frac{p_i}{p_j}$
HD0 of demand:	$\sum_j \frac{\partial x_i^*}{\partial p_j} p_j + \frac{\partial x_i^*}{\partial I} I = 0$	Envelope:	$M'(a) = f_a(x, a) _{x=x^*(a)}$
Elasticity form:	$\sum_j \varepsilon_{ij} + \eta_i = 0$	Roy:	$x_i^* = -\frac{\partial v / \partial p_i}{\partial v / \partial I}$
Cobb-Douglas:	$x_1^* = \frac{\alpha I}{p_1}$		$x_2^* = \frac{(1 - \alpha)I}{p_2}$

9.3 Error checks before submitting

1. Scale (p, I) by t . Demand and v must not move.
2. Add up $p \cdot x^*$. It must equal I whenever u is strictly increasing.
3. Any multiplicative constant or monotone transformation of u must have vanished from the demands.
4. Check $\lambda^* > 0$. Zero or negative means something is wrong.
5. A negative quantity means you assumed interiority when the answer is a corner. Redo with the full Kuhn-Tucker conditions.
6. Count the terms in any Euler sum: $n + 1$ for demand in (p, I) , n for a function of quantities alone.

9.4 Counterexamples to have ready

To break	Use
uniqueness	perfect substitutes, $u = x_1 + x_2$
continuity of demand	the same, via Berge
that income is always spent	bliss point, $-(x - x_0)^2 - (y - y_0)^2$
interiority	$u = x_1$, or quasi-linear with $I \leq p_1$
diminishing MRS giving quasi-concavity	$u = x_1 + (x_2 x_3)^{2/3}$
existence of a utility representation	lexicographic preferences
convexity of preferences	a trip split between two destinations
strict quasi-concavity being necessary	Leontief, unique solution at the kink